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APRIGLIANO
INTERNATIONAL LAW FIRM

PROPOSED MSc THESIS & CURRICULAR INTERNSHIP

Knowledge Systems & AI Legal Automation

Candidate: Nafis Ismam · MSc Computer Science and Engineering, Politecnico di Milano
Company: Docitalis | **Duration:** Six months, full-time

Challenge

Can complex legal reasoning and pleading composition be transformed into a controlled, auditable software system without delegating legal strategy to a generative model?

Core workflow

Case Facts → Legal Strategy → Defences/Sub-defences → Composition Rules → Legal Modules → Complete Pleading

Proposed internship activities

• Work with lawyers to formalise facts, strategies, defences and reusable argumentative modules.

• Model paragraph-selection rules, dependencies, incompatibilities and precise ordering.

• Develop a versioned legal-module knowledge base and deterministic document-assembly engine.

• Implement regression tests against validated historical cases.

• Build a separate AI/RAG review layer connected to a structured Legal Sources Matrix to flag omissions, inconsistencies, citation issues and relevant authorities.

• Identify modules potentially affected by new judgments for lawyer review or update.

• Explore appeals and structured first-instance judgment analysis as a possible later extension.

Potential thesis focus and evaluation

Possible directions include reproducibility and traceability of rule-based assembly, or evaluation of source-grounded AI review. Evaluation could examine agreement with lawyer-validated expected outputs, constraint enforcement, regression behaviour, citation support and accuracy of review alerts, subject to access to suitable anonymised cases.

Technical areas

Python · Data modelling · Knowledge representation · Rule-based systems · Backend development · Software testing · NLP/document processing · Information retrieval · LLM APIs and RAG

The research question, methodology and scope will be agreed with the academic supervisor and company. This document describes proposed work, not completed results.